International Database: World Population Estimates and Projections

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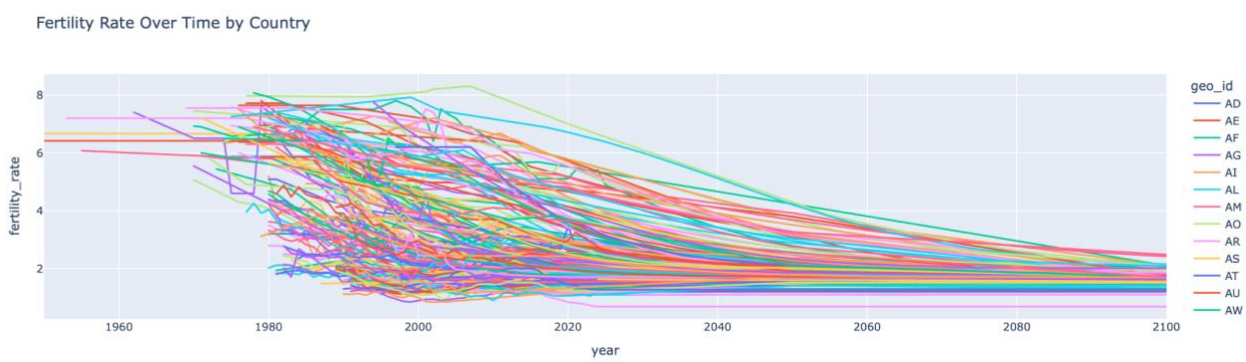
Dataset:

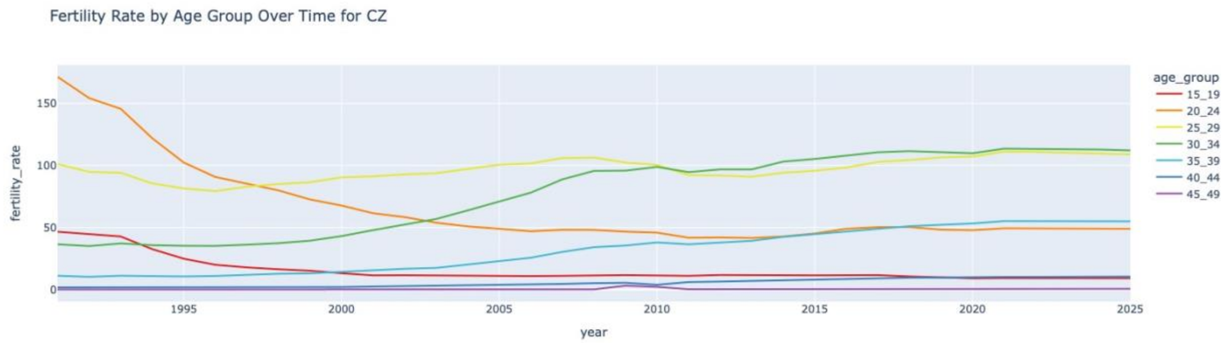
United States Census Bureau. (2024, October 15). International Database (IDB). Retrieved from: https://www.census.gov/programs-surveys/international-programs/about/idb.html

Discoveries:

This project aimed to look for correlations related to fertility rates around the world and see what factors are most closely associated to the fertility rate of a given country. I also compared these factors to the fertility rates of specific age groups so see what impacts the fertility rate of different age groups. The first thing I wanted to look at was how fertility rates were changing over time.

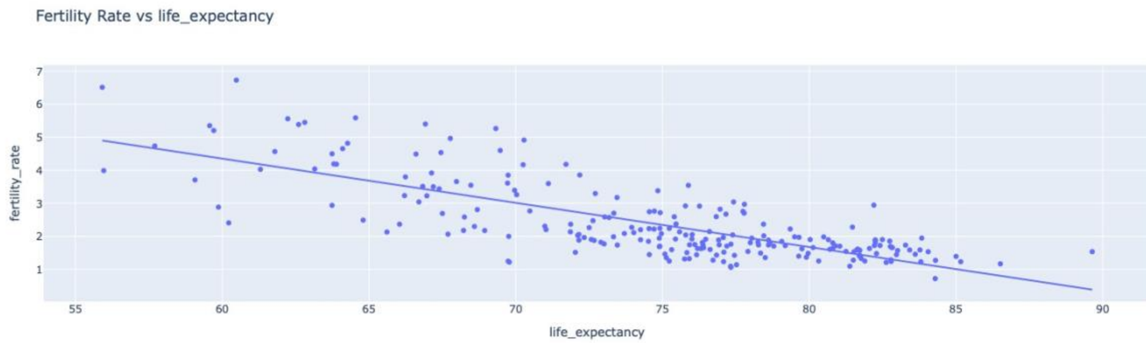


Each line in the above graph shows the fertility rate for each country in the world. It look like a mess but you can see that the overall trend is that the fertility rate around the world is going down and is projected to continue to go down. If you break it down by age group there is another trend that is starting to appear.

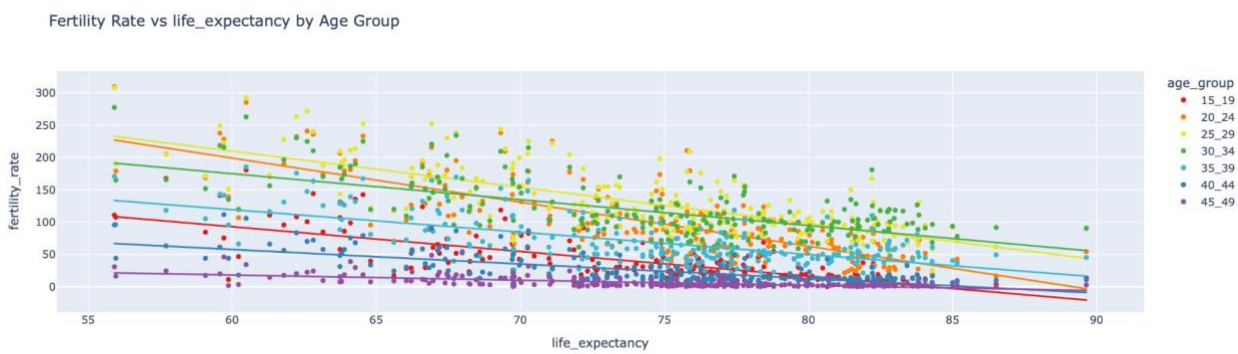


Here is the fertility rate in each age group for Czechia (You might notice that the number on the y-axis are mich higher, that is because the age specific fertility rates are calculated differently, instead of it being births per female, it is the number of births during a year per 1,000 females in that age group). Over time the age group with the highest fertility rate goes from being 20-24 to 25-29 to 30-34 showing that people are deciding to have children at an older age. This is true for most countries around the world although there are still a good portion of countries where the age that people are having children had remained roughly constant over time.

Now its time to see what factors are most closely correlated with fertility rate. I compared fertility rate to life expectancy, infant mortality, population density, sex ratio, and dependency ratio. I hypothosized that life expectancy would be the biggest factor affecting fertility rate because both life expectancy and fertility rate have changed in roughly a consistent direction over time. I decided to use data for the year 2023 because that was the last year before the database was updated and everything after that was just projections.



Here is an example of my scatter plot for fertility rate vs life expectancy for the year 2023, each dot represents a country and there is a clear negative correlation. We can see the strength of the correlation by looking at the line's R-squared value.



Here is the same graph but split by age groups, all seven age groups have a negative correlation but some age groups have a stronger correlation than others. After going through all the data I end up with the following correlation matrix:

	Overall	15-19	20-24	25-29	30-34	35-39	40-44	45-49
Life Expectancy	0.588826	0.559049	0.635312	0.539663	0.389681	0.417975	0.486361	0.418026
Infant Mortality	0.685866	0.625622	0.663687	0.613881	0.527904	0.521747	0.585645	0.523765
Population Density	0.015867	0.015996	0.019764	0.019123	0.009081	0.007436	0.009199	0.007083
Sex Ratio	0.01077	0.011108	0.008045	0.010834	0.01591	0.008661	0.002843	0.000967
Dependency Ratio	0.72937	0.555623	0.581323	0.654358	0.695911	0.698703	0.645622	0.480548

At first glance it becomes clear that population density and sex ratio are not correlated with fertility rate, this surprised me a bit as I had assumed that countries with a lower population density would have a higher fertility rate than countries with a higher population density. That leaves life expectancy, infant mortality, and dependency ratio as the remaining factors that are correlated with fertility rate.

At first glance, it seems that dependency ratio has the biggest impact on fertility rate but upon close thought, that may not be the case. Dependency ratio is the ratio of people under 18 and above retirement age in the overall population and it would make sense that there would be a strong positive correlation with fertility rate. Countries with a higher fertility rate will have more children and thus a higher dependency ratio, basically the fertility rate is affecting the dependency ratio, not the other way around.

That leaves infant mortality as the biggest factor affecting fertility rates. This makes perfect sense as in societies where the survival of a child is uncertain, parents would be inclined to have more children to ensure that some make it to adulthood. Unlike in societies where children are expected to make it to adulthood, there is less of an incentive to do that because you already expect the children you have to make it to adulthood.

What can be improved:

The data analyzed still leaves many questions unanswered. While infant mortality was the biggest facter affecting fertility rates compared to the other factors analyzed, there are still many other factors to consider such as education, socioeconomic status, human development, culture, and more. None of the factors analyzed have also been able to explain the trend of people having children at an older age. Moving forward, data from additional datasets can be included to give us greater insights and hopefully answer these additional questions.